

Robot-Assisted Stone Dusting: A Bench-Top Study

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Abstract

This study evaluates whether robot-assisted control of the fiber-to-stone standoff distance (SD) can improve laser stone dusting under bench-top conditions. A Dobot Nova 5 robot was used to position a Ho:YAG laser fiber above submerged BegoStone samples. Fixed-SD and dynamic-SD strategies were compared using OCT-based crater measurements, including crater volume, average depth, and profile area. The results show that dynamic SD adjustment can affect crater morphology and improve ablation performance under selected non-optimal initial SD conditions. These findings support the potential value of robotic SD control in laser lithotripsy.

Keywords: robot-assisted lithotripsy, laser stone dusting, OCT, crater analysis

1 Background and Introduction

Laser lithotripsy is an important approach for the treatment of urinary stones. Among its various techniques, stone dusting performed with low pulse energy and high repetition rate has attracted broad attention because it can gradually reduce stones to fine particles and decrease the need for fragment extraction [1, 3]. However, in practical procedures, conventional manual dusting relies heavily on the surgeon's experience to maintain the relative position between the laser fiber and the stone surface. This process is labor-intensive and makes it difficult to preserve an effective fiber-to-stone standoff distance (SD) during repeated laser pulses, thereby limiting energy delivery efficiency and the consistency of stone ablation. A more controllable and efficient dusting strategy is therefore needed.

Although the importance of SD has been established, most previous studies have evaluated ablation performance under fixed-SD conditions [2,4]. In actual stone dusting, however, crater formation continuously changes the fiber-to-stone geometry as ablation proceeds. As a result, an initially favorable SD may no longer remain optimal throughout treatment. This limitation makes it difficult to determine whether maintaining a constant SD is sufficient, or whether active SD adjustment during treatment could provide a measurable advantage.

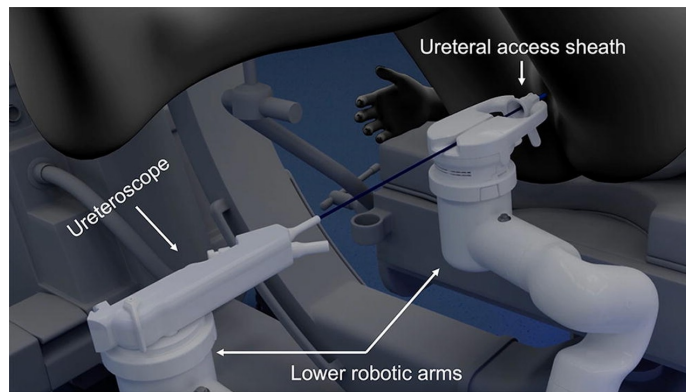


Figure 1: Schematic illustration of a robot-assisted flexible ureteroscopy system for laser lithotripsy.

This gap motivates the present study. In this work, a robot-assisted bench-top platform was used to investigate whether dynamic fiber-to-stone SD control can improve stone dusting efficiency and modulate crater morphology compared with fixed-SD strategies. By combining robotic pulse-level position control with optical coherence tomography (OCT)-based crater measurement, this study evaluates crater volume, average depth, and profile area under different SD control conditions. The results are intended to clarify whether dynamic SD adjustment provides advantages over fixed-distance treatment as crater geometry evolves.

2 Specific Aims and Significance

This study aims to investigate whether dynamic fiber-to-stone standoff-distance (SD) control can improve stone dusting efficiency and modulate crater morphology in robot-assisted laser lithotripsy. To address this overall objective, three specific aims were defined. First, this work seeks to achieve precise SD control between the laser fiber and the stone surface using a robotic system and to evaluate whether the robot can perform pulse-level position adjustment with a speed and repeatability that are difficult to achieve manually. Second, under bench-top experimental conditions, this study systematically compares fixed-SD and dynamic-SD strategies to determine whether dynamic control can improve ablation efficiency under non-optimal initial SD conditions. Third, by analyzing crater volume, average depth, and profile area under different SD control strategies, this work examines whether robotic SD control can serve as an effective method for manipulating damage geometry.

This study provides bench-top evidence for whether robotic pulse-level SD control improves ablation efficiency and modulates crater morphology compared with fixed-SD strategies.

3 Materials and Method

3.1 Robot-Assisted Stone Dusting Setup

In this study, a robot-assisted laser lithotripsy bench-top platform was developed to achieve precise relative positioning between the laser fiber and the stone surface under controlled conditions. The platform mainly consisted of a Dobot Nova 5 robotic arm, a Ho:YAG laser fiber mounted on the robot end-effector, a BegoStone sample submerged in a water tank, and an imaging system for experimental observation and post-treatment measurement. The robotic arm was used to control the spatial position of the laser fiber during the dusting process, thereby enabling precise regulation of the fiber-to-stone standoff distance (SD).

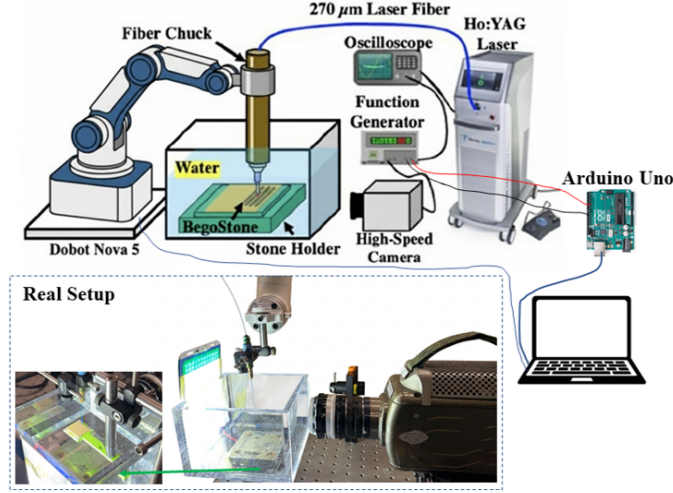


Figure 2: Experimental setup of the robot-assisted stone dusting system.

In the experimental setup, the BegoStone sample was fixed and immersed in water to simulate the fluid environment of laser lithotripsy. The laser fiber was positioned perpendicular to the stone surface to minimize the influence of incidence angle variation on the results and to ensure good comparability across experimental groups. The fiber diameter used in the experiments was $365\ \mu\text{m}$, and the BegoStone samples were prepared using a powder-to-water ratio of 5:2. In addition, a high-speed camera was placed laterally to record bubble dynamics and crater formation during laser firing, while optical coherence tomography (OCT) was used after treatment to perform three-dimensional scanning of the ablation craters and obtain quantitative metrics including crater volume, average depth, and profile area.

3.2 Laser Parameters and Experimental Conditions

To ensure comparability across different treatment strategies, all experiments were conducted under the same laser settings and environmental conditions. A Ho:YAG laser system was used, with a pulse energy of $0.4\ \text{J}$ and a repetition rate of $20\ \text{Hz}$, and all treatments were performed in water. The laser fiber had a diameter of $365\ \mu\text{m}$ and was kept perpendicular to the stone surface in order to minimize the influence of incidence angle variation on ablation outcomes. BegoStone samples were prepared using a powder-to-water ratio of 5:2 to simulate the properties of urinary stones.

In the systematic experimental design, the total number of pulses was set to $N \in \{20, 40, 80, 160, 320, 640\}$, and the initial standoff distance was set to $SD_0 \in \{0, 0.5, 0.75, 1.0\}\ \text{mm}$. The fixed-SD strategy served as the baseline condition, in which the robot maintained a constant standoff distance between the fiber tip and the stone surface throughout the entire pulse sequence. In contrast, the dynamic-SD strategy adjusted the fiber position stepwise during laser firing to evaluate how different initial SD values and their dynamic variations affected ablation efficiency and damage morphology. Each strategy was typically repeated 5 to 8 times, and adjacent craters were spaced approximately $4\ \text{mm}$ apart to reduce interference.

Table 1: Experimental conditions and variables.

Parameter	Value
Laser type	Ho:YAG
Pulse energy	0.4 J
Pulse repetition rate	20 Hz
Environment	In water
Fiber orientation	Perpendicular to top surface
Total pulse number, N	{20, 40, 80, 160, 320, 640}
Initial SD, SD_0	{0, 0.5, 0.75, 1.0} mm

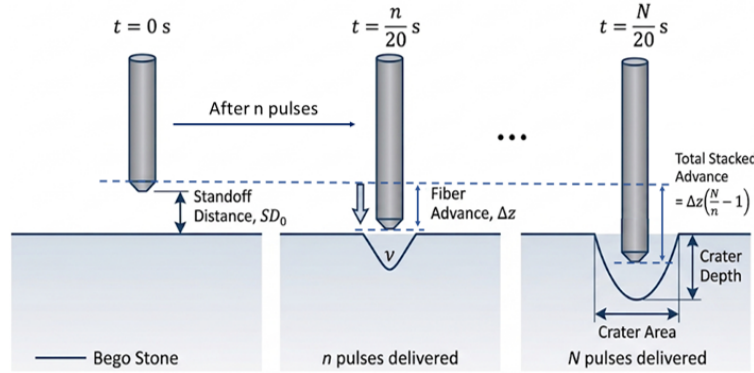
3.3 Standoff Distance Control Strategies

In this study, both fixed standoff-distance (SD) control and dynamic SD control strategies were designed to compare how different fiber-positioning schemes affect laser stone dusting performance. The fixed-SD strategy served as the baseline condition, in which the relative distance between the laser fiber tip and the stone surface was kept constant throughout the entire pulse sequence. In contrast, the dynamic-SD strategy periodically adjusted the robot end-effector position during laser firing so that the fiber-to-stone distance changed stepwise as ablation progressed.

For the dynamic-SD strategy, the robot updated its position once every n laser pulses. If the total number of pulses was N , the number of SD adjustment steps can be expressed as

$$K = \frac{N}{n}.$$

At each update, the robot incrementally translated the fiber tip along the fiber axis by a displacement denoted as Δz . In this way, the fiber was moved downward or upward in discrete steps to accommodate the continuous evolution of crater morphology during ablation. According to the experimental design, the update interval was $0.05n$ s, and the programmed robot motion speed was 16 mm/s.

**Figure 3:** Dynamic standoff-distance control strategies.

In the systematic experiments, the dynamic strategy was mainly used to evaluate whether dusting efficiency could be improved by transitioning from a non-optimal initial SD toward a more favorable SD condition. Specifically, for several non-optimal initial distances, the robot gradually adjusted the fiber position toward $SD = 0.5$ mm to compare the performance of dynamic adjustment against that of fixed-SD strategies. Under selected conditions, multi-step strategies with different update frequencies, such as 2-step

and 4-step protocols, were also compared to assess whether more frequent SD updates could better track crater evolution and maintain effective energy coupling.

3.4 Experimental Procedure

Before each experiment, the position of the laser fiber tip was first calibrated according to the prescribed initial standoff distance (SD), and the laser parameters were then configured. The robot subsequently carried out the stone dusting experiment according to the predefined control strategy, including either the fixed-SD strategy or the dynamic-SD control strategy. Under the dynamic strategy, the robotic arm adjusted the fiber position stepwise based on the preset pulse interval and displacement increment, thereby enabling real-time regulation of the fiber-to-stone relative distance.

For each experimental condition, multiple independent craters were created on the surface of the Bego-Stone sample as repeated trials. Adjacent craters were spaced approximately 4 mm apart to minimize interference between neighboring ablation regions. Each strategy was typically repeated 5–8 times to reduce the influence of experimental variability and improve the reliability of comparisons across groups.

After laser treatment, the sample surface was first dried, and each crater was then imaged in three dimensions using optical coherence tomography (OCT). The resulting OCT data were subsequently used for crater segmentation and quantitative analysis to extract geometric metrics including crater volume, average depth, and profile area.

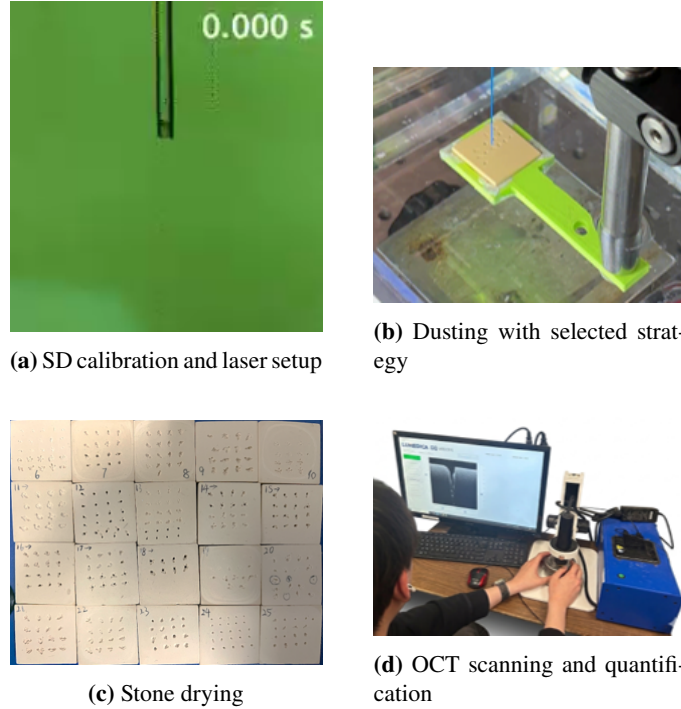


Figure 4: Bench-top experiment workflow.

3.5 OCT Measurement and Quantitative Metrics

After laser treatment, optical coherence tomography (OCT) was used to measure the three-dimensional geometry of each crater. In this study, three quantitative metrics were extracted, namely crater volume, average

depth, and crater profile area, to characterize the differences in ablation performance under different standoff-distance control strategies. The OCT system used in this study provided a resolution of $512 \times 512 \times 512$ voxels, with a voxel size of $15.1 \times 15.1 \times 4.65 \mu\text{m}^3$.

Among these metrics, crater volume was calculated from the number of voxels within the segmented crater region and can be expressed as

$$V = N_{\text{crater}} \Delta x \Delta y \Delta z,$$

where N_{crater} denotes the total number of voxels inside the crater region, and Δx , Δy , and Δz denote the voxel dimensions along the three axes. Maximum crater depth was defined as the greatest distance between the local surface reference and the deepest segmented crater boundary. Crater profile area was used as an important geometric metric to characterize the planar extent of the crater opening. Together with crater volume and depth, it was used to comprehensively evaluate ablation morphology across different experimental groups.

These three metrics enabled quantitative comparison of crater geometry in terms of volumetric expansion, axial penetration, and lateral morphological variation, thereby providing a basis for evaluating the effects of fixed-SD and dynamic-SD control strategies on laser stone dusting performance.

4 Results

4.1 Baseline Results Under Fixed Standoff-Distance Conditions

We first evaluated crater volume under fixed-SD conditions to identify a reference distance for subsequent dynamic-control experiments. As shown in Fig. 5, crater volume increased with pulse number under all four fixed-SD conditions, but the $\text{SD} = 0.5 \text{ mm}$ group consistently produced the largest crater volume across the tested pulse range. In contrast, $\text{SD} = 0 \text{ mm}$ yielded the lowest volume, while $\text{SD} = 0.75 \text{ mm}$ and $\text{SD} = 1.0 \text{ mm}$ showed intermediate performance. Based on these baseline results, $\text{SD} = 0.5 \text{ mm}$ was selected as the reference condition for subsequent dynamic-SD comparisons.

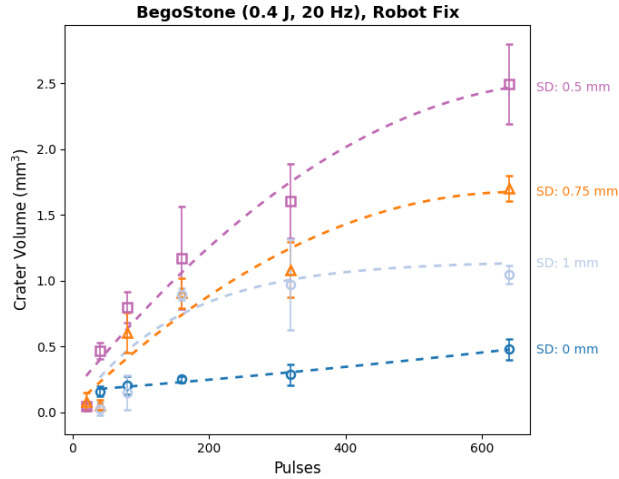


Figure 5: Fixed SD baseline.

4.2 Dynamic Adjustment from Non-Optimal SD to the Optimal SD

Based on the fixed-SD baseline results, we next examined whether gradually adjusting the fiber position toward $\text{SD} = 0.5 \text{ mm}$ during ablation could improve crater volume when treatment started from non-optimal initial SD conditions. As shown in Fig. 6, the effect of dynamic SD adjustment depended on the initial SD.

Among the three transition cases, namely 0 mm to 0.5 mm, 0.75 mm to 0.5 mm, and 1.0 mm to 0.5 mm, the clearest improvement was observed when the initial SD was 0.75 mm. Improvement was also observed in the 0 mm to 0.5 mm and 1.0 mm to 0.5 mm cases, but the separation between dynamic and fixed conditions was less pronounced. These results indicate that the benefit of dynamic SD control was condition-dependent rather than uniform across all non-optimal starting distances.

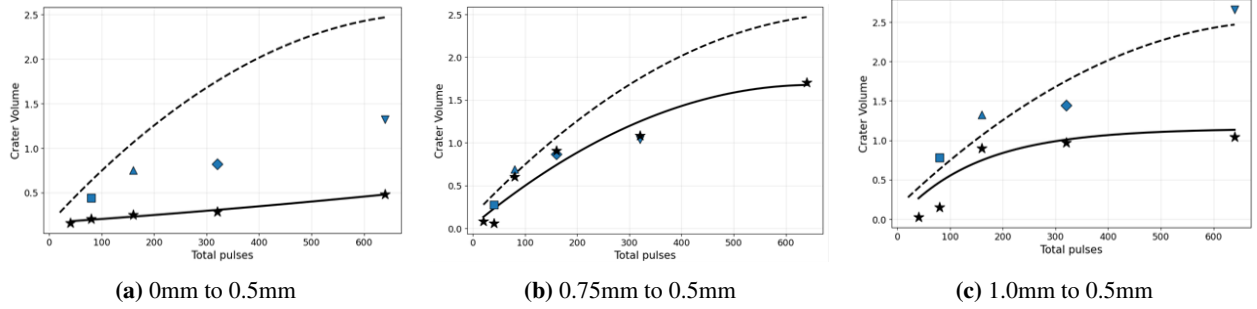


Figure 6: Crater volume results.

4.3 Optimal Enhancement Under the Initial SD=0.75 mm Condition

Among the tested non-optimal initial SD conditions, the initial SD = 0.75 mm case showed the clearest benefit from dynamic control. As shown in Fig. 7, the 4-step strategy produced the highest crater volume under this starting condition and outperformed both the SD = 0.75 mm fixed and 2-step strategies. At 80, 160, and 320 pulses, the crater volume under the 4-step strategy increased by 34.4%, 27.1%, and 44.5%, respectively, relative to the SD = 0.75 mm fixed condition. Further statistical analysis showed that these improvements were significant, with $p = 1.89 \times 10^{-2}$, 2.57×10^{-2} , and 1.73×10^{-3} compared with the SD = 0.75 mm fixed strategy, and $p = 3.17 \times 10^{-2}$, 3.18×10^{-2} , and 1.56×10^{-3} compared with the 2-step strategy, respectively. In contrast, the 2-step strategy showed only limited or inconsistent improvement. These results indicate that, under the initial SD = 0.75 mm condition, more frequent stepwise adjustment provided the most evident volume enhancement.

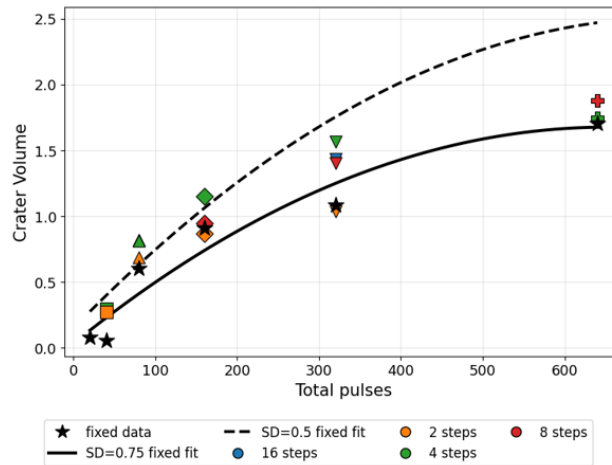


Figure 7: Comparison of crater volume under different step control strategies for the initial SD = 0.75 mm condition.

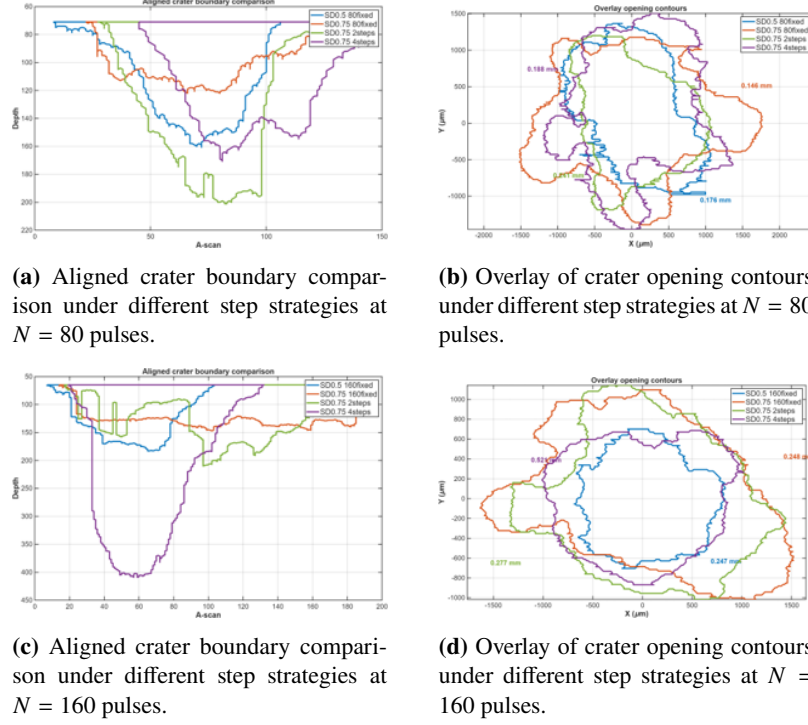


Figure 8: Comparison of crater morphology under different step strategies at $N = 80$ and 160 pulses.

4.4 Limited Enhancement Under the Initial $SD=0.5$ mm Condition

In contrast, when the initial SD was already near the favorable condition ($SD = 0.5$ mm), dynamic updating did not provide a consistent advantage. As shown in Fig. 9, crater volume under stepwise adjustment was not systematically higher than that of the fixed- SD condition, and some dynamic strategies produced lower volumes at selected pulse numbers. The representative morphologies in Fig. 10 further show that the differences among strategies at this initial SD were less favorable than those observed for the initial $SD = 0.75$ mm condition. These results show that dynamic stepwise updating did not provide a consistent quantitative advantage when the initial SD was 0.5 mm.

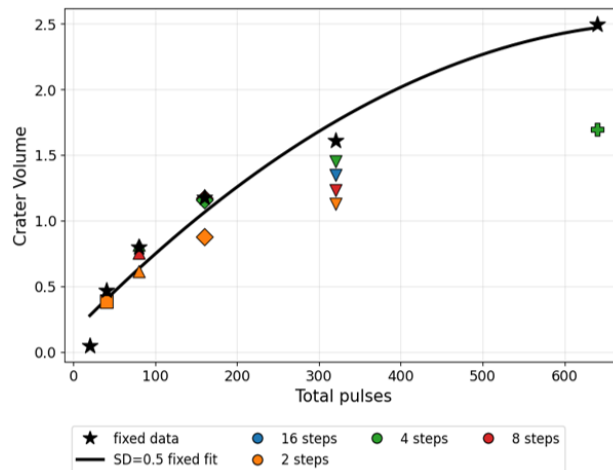
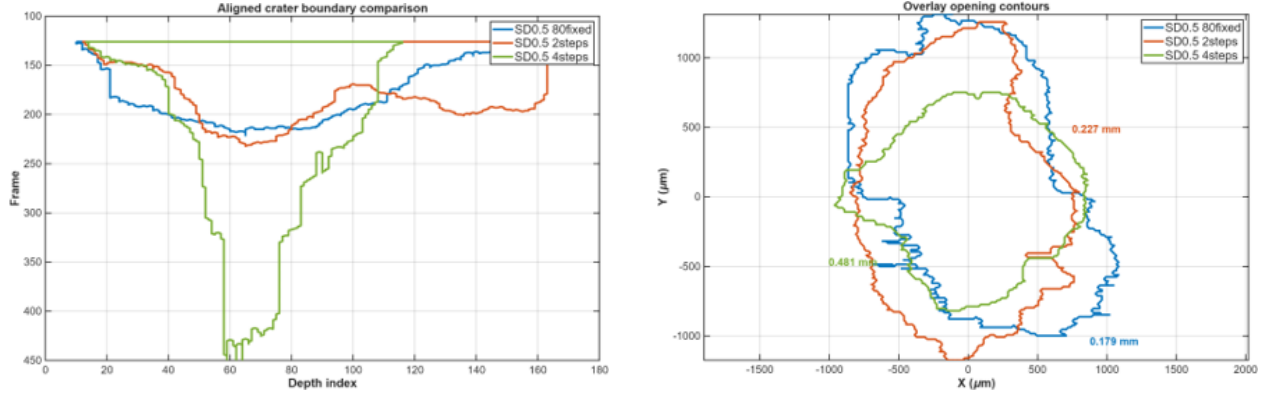


Figure 9: Comparison of crater volume under different step strategies for the initial $SD = 0.5$ mm condition.



(a) Aligned crater boundary comparison under different step strategies at $N = 80$ pulses.

(b) Overlay of crater opening contours under different step strategies at $N = 80$ pulses.

Figure 10: Comparison of crater morphology under different step strategies at $N = 80$.

4.5 Analysis of the Main Source of Volume Enhancement

To further examine the main source of volume enhancement under dynamic SD control, we compared sample-level crater volume change with both average depth change and profile area change under different initial SD conditions. As shown in Fig. 11, crater volume change was more strongly associated with profile area change than with average depth change. Representative crater morphologies further showed that different SD control strategies produced distinct crater shapes.

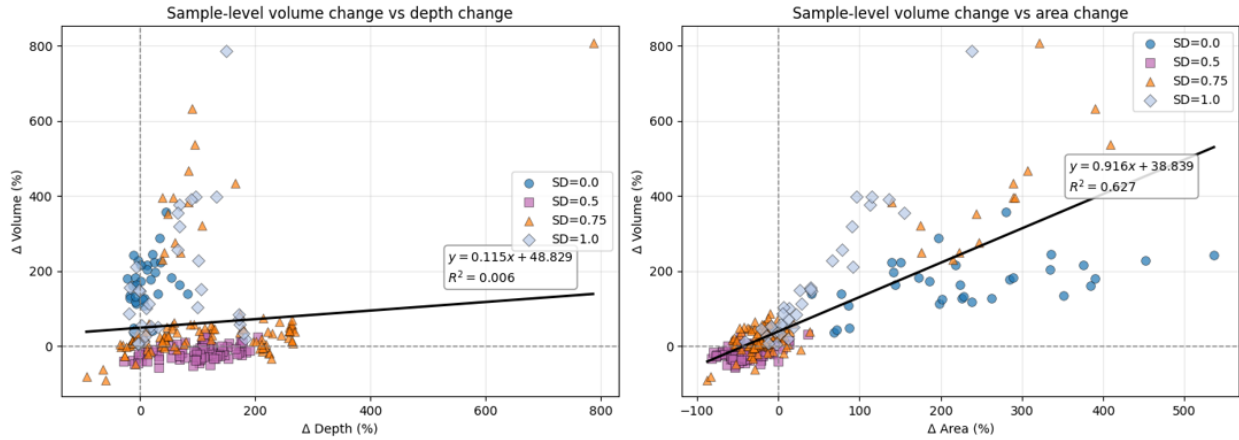


Figure 11: Comparison of crater volume change versus average depth change and profile area change under different initial SD conditions.

5 Discussion and Summary

This study evaluated the effects of dynamic fiber-to-stone standoff-distance (SD) control on stone dusting efficiency and crater morphology using a robot-assisted laser lithotripsy bench-top platform. Under fixed-SD conditions, 0.5 mm produced the best ablation volume. For non-optimal initial SD conditions, gradually adjusting the fiber position toward 0.5 mm improved ablation performance, with the clearest enhancement observed for the 4-step strategy when the initial SD=0.75 mm.

Further analysis showed that the volume improvement from dynamic SD control was driven mainly by an increase in crater profile area rather than by a substantial increase in average depth, suggesting an important role of cavitation-related effects and energy coupling [1]. Different SD strategies also altered crater geometry: larger SD values tended to produce broader, shallower damage, whereas smaller SD values produced more localized and deeper damage.

The robotic system enabled precise pulse-level SD control with speed and repeatability beyond manual operation. Dynamic SD modulation improved dusting efficiency under non-optimal conditions and provided a simple way to manipulate damage morphology. Rather than only identifying a fixed optimal SD, this study shows that active SD adjustment during crater evolution may be more effective than maintaining a constant distance. This idea extends prior work that demonstrated strong dependence of dusting efficiency on standoff distance under non-optimal initial SD conditions [4].

Future work should focus on improved control strategies, broader experimental validation, and system integration. For the limited gain observed at SD=0.5 mm, combining axial SD adjustment with lateral scanning may be more effective than depth tracking alone [5]. Real-time adaptive control based on crater evolution should also be explored. Additional validation under more clinically relevant conditions, including irregular stone surfaces, is needed, together with more repeated trials and re-testing of manual fixed and robot fixed baselines on the same platform. Other robotic variables, such as fiber angle and more complex motion strategies, also warrant further investigation.

Acknowledgment

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References

- [1] J. Chen, D. S. Ho, G. Xiang, G. Sankin, G. M. Preminger, M. E. Lipkin, and P. Zhong, “Cavitation plays a vital role in stone dusting during short pulse holmium:YAG laser lithotripsy,” *Journal of Endourology*, vol. 36, no. 5, pp. 674–683, 2022. doi: 10.1089/end.2021.0526.
- [2] A. H. Aldoukhi, W. W. Roberts, T. L. Hall, and K. R. Ghani, “Watch your distance: The role of laser fiber working distance on fragmentation when altering pulse width or modulation,” *Journal of Endourology*, vol. 33, no. 2, pp. 120–126, 2019. doi: 10.1089/end.2018.0572.
- [3] A. H. Aldoukhi, K. M. Black, T. L. Hall, W. W. Roberts, and K. R. Ghani, “Frequency threshold for ablation during holmium laser lithotripsy: How high can you go?” *Journal of Endourology*, vol. 34, no. 10, pp. 1075–1081, 2020. doi: 10.1089/end.2020.0149.
- [4] J. Chen, D. Li, W. Yu, Z. Ma, C. Li, G. Xiang, Y. Wu, J. Yao, and P. Zhong, “The effects of scanning speed and standoff distance of the fiber on dusting efficiency during short pulse holmium:YAG laser lithotripsy,” *Journal of Clinical Medicine*, vol. 11, p. 5048, 2022. doi: 10.3390/jcm11175048.
- [5] F. Panthier, T. Germain, C. Gorny, L. Berthe, S. Doizi, and O. Traxer, “Laser fiber displacement velocity during Tm-fiber and Ho:YAG laser lithotripsy: Introducing the concept of optimal displacement velocity,” *Journal of Clinical Medicine*, vol. 11, Art. no. 181, 2022. doi: 10.3390/jcm11010181.